

Ganges Water Level Prediction using Machine Learning Methods

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Ganges Water Level Prediction using Machine Learning Methods

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DECLARATION

I certify that

- A. The work contained in this report is original and has been done by me under the guidance of my supervisor.
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This is to certify that the Dissertation Report entitled, “Ganges Water Level Prediction using Machine Learning Methods” submitted by Mr. Lakshya Sharma to Indian Institute of Technology, Kharagpur, India, is a record of Bonafide Project work carried out by him under my supervision and guidance and is worthy of consideration for the award of the degree of Master of Technology in Ocean Engineering and Naval Architecture of the Institute.

Supervisor

Date:

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Abstract

For the best management of various water resource projects, accurate water level (WL) prediction is required. In the field of water resources management, developing a reliable model for WL forecast remains a difficult task. The government is trying to replace the fixed pontoon bridges over Ganges by the floating barges with a connection-disconnection system with a dynamic positioning system being designed. Forecasted water levels can be used to manage these pontoon kulfi systems with mechanical opening mechanisms effectively. Hence, improving the navigation channel for better transportation of goods by National Waterway 1.

Machine Learning based models ARIMA and LSTMs were proposed in this study to forecast the average monthly and daily WL of Ganges at Chak, Bihar using satellite data. One year of data was used for testing the quality of the model. RMSE was used as a metric to select the best possible model. In conclusion, the results show that the ML approaches are suitable in Water Level prediction, and may serve as robust and trustworthy forecast tools for the research region.

Chapter 1. Introduction

Settlements were built on the banks of rivers in ancient times with the aim that their occupants could benefit from the river's trade, food, and defense options, and exactly the same applies in modern times too. Water is required for human survival. The household, industrial, agricultural, and energy sectors all rely on river water. As a result, understanding the anticipated behavior of rivers becomes critical for engineers.

The Ganges is an Asian trans-border river that runs through India and Bangladesh. The river has a length of 2,525 kilometers and originates in the western Himalayas of Uttarakhand, India. The Ganges' main stem begins in Devprayag, at the conflux of the Alaknanda, which is considered the source river in hydrology due to its larger length.

It travels south and east through North India's Gangetic plain, receiving the Yamuna as a right-bank lower stream and many left-bank streams from Nepal, which makes for the majority of its flow. The Ganges is renamed the Padma as it flows into Bangladesh. It is then confluent with the Jamuna, the Brahmaputra's tributary, and the Meghna, forming the Ganges Delta's primary estuary and flowing into the Bay of Bengal.

For billions of people who reside along the Ganges basin and rely on it for their day to day requirements, the Ganges is a lifeline. Many ancient provincial or imperial capitals, including Pataliputra, Kashi, Hajipur, Patna, Delhi and Kolkata, are situated at its banks or on the banks of its lower streams and associated canals. Around 140 species of fish, 90 species of amphibians, reptiles, and mammals call

the river home, including severely endangered species like the gharial and South Asian river dolphin. The Ganges has been the most sacred source of water to Hindus.



Fig. 1 : Journey of Ganges from its origin to Bay of Bengal

The behavior of river water is frequently unpredictable and unexplained. There are basically two types of approaches that can be used to study and control water behavior:- Structural and Non-structural. Structural approach includes making reservoirs, dams, and barrages while Non-structural approach involves disaster management, disaster response systems, and floodproofing. Hidden information, such as the flow of water at a certain period, can be exposed using predicting algorithms based on previous values, which can aid quick emergency actions and minimize tragedies. Forecasting water levels is a non-structural type metric and is necessary for improved water resource management. The ability to accurately forecast a river's water level enhances flood control systems and could serve as a

precautionary signal for rapid decision-making and planning to mitigate the effects of floods, which are regarded as one of the most dangerous natural risks to life and property. The forecast can also be used to manage some floating barges which is also the main objective of the project. The Ganges has been used for navigation and transportation of people and goods for thousands of years. The Ganga-Bhagirathi-Hooghly river system, also known as NW-1, spans across the Ganges river from Prayagraj in Uttar Pradesh to Haldia in West Bengal, via Patna and Bhagalpur in Bihar. It is India's longest waterway, measuring 1,620 kilometers (1,010 miles) in length. Because of its strategic location, it is the most important of all the national waterways. The NW-1 caters the day to day needs of major towns and their industrial centers in Bihar, West Bengal, Jharkhand, and Uttar Pradesh.

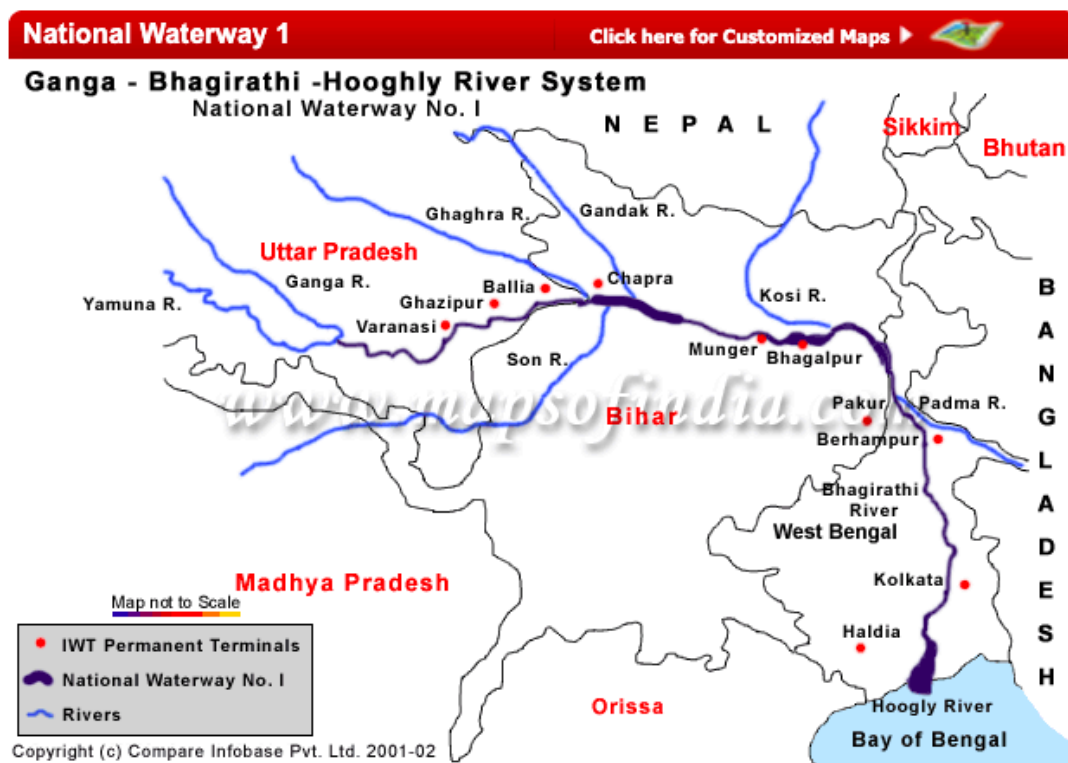


Fig. 2 : National Waterway No 1

With the technical and financial aid of the World Bank, the government has started work on the Jal Marg Vikas Project (JMVP) for capacity augmentation of navigation on National Waterway-1 (NW-1) on the Haldia – Varanasi segment of the Ganga-Bhagirathi-Hooghly River System. The project's main features include the construction of multimodal terminals in Varanasi, Sahibganj, and Haldia, Ro-Ro terminals, a navigational lock in Farakka, channel marking systems, integrated vessel repair and maintenance facilities, automated information techniques such as the Differential Global Positioning System (DGPS), slipways, and River Information System (RIS), bunkering facilities, day and night navigation aids, river training, and river navigation aids. On the ground, projects worth approximately Rs. 1800 crore have begun in just three years after legal clearances.

Chapter 2. Application and Motivation

One of the major issues in the JMVP is the existence of pontoon bridges which presents significant challenges to navigation. Presently there are more than 13 pontoon bridges over Ganga river over proposed waterway 1. These are fixed and welded structures. So, Everytime when a barge has to pass, these are cut and welded after it crosses the bridge. Therefore the government is trying to replace these with a floating barge with a connection-disconnection system with a dynamic positioning system being designed. Forecasted water levels can be used to manage these pontoon kulfi systems with mechanical opening mechanisms effectively.

This research project tries to forecast the water level of the Ganges River recognizing the significance of the preceding debate. The purpose of this study is to provide an improved prediction tool for predicting water levels based on the Automatic ARIMA model, which can accurately anticipate and analyze various water level patterns throughout time. The accomplished goal will be considerable to the relevant authorities as it will provide them an opportunity of efficiently managing the new pontoon kulfi system which will help in improving the navigation system along NW 1.

Chapter 3. Literature Review

River Water Level forecasting has been the theme of various studies all over the world. Previously, traditional methods were employed to forecast capacity, overflow discharge, and water level; but, Machine Learning approaches (ML) are now rapidly being applied in forecasting [1,2]. The term machine learning (ML) refers to the ability of machines to evaluate, cluster, extract complicated links, and make judgments without the need for programming. A further feature of ML is its capability to detect patterns in input data and deliver results by analyzing the data's hidden complex structures .

Probability characteristics, time series approaches, artificial data generation, multivariate regressions, pattern identification, and neural network methods are all examples of these techniques. ARIMA models have been deemed one of the best choices in modeling the time series and future forecasting[3].

Only a few attempts to predict the WL using the SARIMA model have been reported in the literature, such as estimates of lake water levels [4] and groundwater levels [5]. The SARIMA model, on the other hand, has the benefit of having minimal model parameters to explain non-stationary time series both within and between seasons. When opposed to Machine Learning (ML) approaches, which frequently require several parameters as input, it is a substantial simplification.

ARIMA is one among the various most effective and accurate linear statistical models for time series forecasting [6]. One of the most well-known statistical time series models for RWL prediction is the autoregressive integrated moving average

(ARIMA) model [7,8]. The use of ARIMA in hydrological studies has been widely documented. Water quality [9], groundwater anomaly [10], runoff, drought, monthly streamflow [11,12], river discharge [13,14], and rainfall have all been successfully analyzed using ARIMA.

Chapter 4. Time Series Analysis Methods

1. ADF Test

The Augmented Dickey Fuller (ADF) test is used to evaluate the stationarity in a time series. The ADF test was developed by David Dickey and Wayne Fuller in 1979. It analyses the time series for the existence of a unit root under the null hypothesis. The following is the mathematical expression for ADF:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \delta_1 \Delta y_{t-1} + \dots + \delta_{p-1} \Delta y_{t-p+1} + \epsilon_t$$

where α : constant

β : coefficient of time trend

p : the lag order

ϵ_t : is the error term

The test for the null hypothesis $\gamma = 0$ is run after selecting the proper number of lags (p). Non-stationary time series can be converted to stationary by regressing or differencing the dataset till it becomes stationary.

2. ARIMA

The idea for the model was first introduced by Norbert Wiener et al. in the 1930s. The model consists of three parts: Autoregressive (AR), Integrated (I), and Moving Average (MA). The AR part of the equation represents the connection between present and prior observations, the MA part represents the autocorrelation component of error, and the I part represents the number of differencing steps of the series necessary to make the series stationary. ARIMA is among the most efficient and accurate linear statistical models for time series forecasting. ARIMA was used for forecasting in 1970 by Box Jenkins. Ever Since, it has been used in various domains, including engineering, hydrology, economics, and social analysis. Peter Whittle proposed the first general form of ARMA in 1951 , which can be represented as:

$$X_t = c + \epsilon_t + \sum_{i=1}^p \phi_i X_{t-i} + \sum_{i=1}^q \theta_i \epsilon_{t-i}$$

where ϵ_t : white noise term

ϕ, θ : coefficients of the time series.

The mathematical representation of AR (p) and MA (q) is given below:

$$y_t = c + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \beta_3 y_{t-3} + \dots + \beta_p y_{t-p} + \epsilon_t$$

This is an instance of multivariable regressions with lagged y_t value as predictor. It's known as AR(p), with p denoting the AR model of order (p)

MA (q), q (number of moving average terms), where d is the number of differentiation steps taken to make the data stationary.

$$y_t = c + \epsilon_t + \alpha_1 \epsilon_{t-1} + \alpha_2 \epsilon_{t-2} + \alpha_3 \epsilon_{t-3} + \dots + \alpha_p \epsilon_{t-p}$$

The time series forecast was performed using an automated ARIMA tool “Auto-Arima”, which enables users to find appropriate ARIMA orders and conduct the forecast. The selection of ML models, as well as their ordering, were identified using the Auto-ARIMA tool, which is not confined to ARIMA modeling but also evaluates a number of modeling processes. From the set of models, the one with the lowest Akaike Information Criteria (AIC) or Bayesian Information Criteria (BIC) was picked.

3. AIC

Hirotsugu Akaike was the first to use AIC in 1971. The prediction error is calculated by the AIC, and it is used to compare the robustness and accuracy of a statistical model to that of other models. Mathematically It can be represented as[50]:

$$AIC = 2k - 2 \ln(L)$$

where k : model's estimated parameters

L : likelihood function's highest value

The model with the lowest AIC is chosen from a set of models based on the goodness-of-fit measure. In addition, the AIC incorporates a penalty structure that prevents overfitting and hence enhances goodness-of-fit.

4. BIC

The Bayesian Information Criteria (BIC) was first published by Gideon E. Schwarz in 1978 [51]. The formula for BIC is somewhat similar to AIC, with the exception of a penalty for a different number of factors. The penalty system in AIC is "k," whereas that in BIC is $\ln k$. Mathematically, BIC may be represented as :

$$BIC = k \ln(n) - 2 \ln(L)$$

where k is the number of parameters estimated by the model, n is the number of data points, and L is the maximized value of the likelihood function of the model.

BIC can also be used as a model selection criterion, with the model corresponding to lowest BIC value being chosen.

5. Auto-ARIMA

The use of the Automated ARIMA tool is motivated by the need to select acceptable values for p (number of autoregressive terms), d (differencing levels), and q (number of moving average terms). The determination of (p,d,q) is a hard and arduous task because the ARIMA method comprises of (p,d,q), however the Auto- ARIMA function will provide the best model orders automatically based on the lowest values of the given parameters like AIC or BIC.

The ARIMA model's automatic model selection specification may be broken down into four steps:

- I. Use unprocessed or altered data, such as the dependent variable's logs.

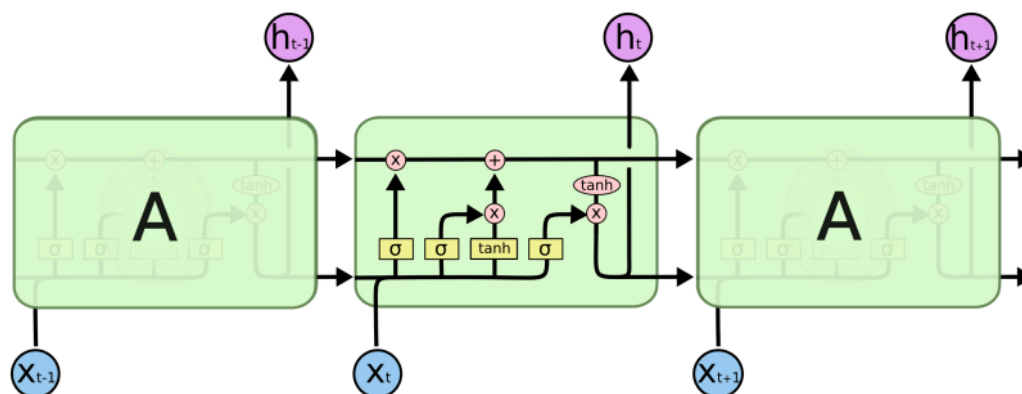
- II. Choosing the right order of integration (I) for the dependent variable.
- III. Exogenic regressors are assessed.
- IV. Using the evaluating technique, choose the ARIMA model's order.

6. LSTMs

Long Short Term Memory networks, or "LSTMs," are a kind of RNNs that can determine long-term relationships. It was firstly introduced by Hochreiter & Schmidhuber in 1997 and several researchers have since then developed and popularized these NNs in subsequent work. Currently, they are widely used and function exceptionally well in several research domains.

LSTMs were created especially to address the issue of long-term reliance. They don't have to try very hard to retain information for long periods of time.

A sequence of repeating neural network modules make up all recurrent neural networks. In conventional RNNs, this repeating module will have a simple network design, e.g a tanh layer. Although LSTMs have a chain-like structure, the recurring module differs. For instance, there are four NN layers instead of one, each of which communicates differently.



The repeating module in an LSTM contains four interacting layers.

7. Error Metrics

The root mean squared error (RMSE), mean absolute percentage error (MAPE), and mean absolute error (MAE) were used to determine error analysis .

The square root of the total of all squares of the differences between expected and actual values is the root of the RMSE. RMSE represents the residual spread range. Mathematically,It can be expressed as :

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (Predicted - Actual)^2}{N}}$$

where N : number of observations

i : variables along with predicted values and actual values

In a forecast study, MAE is just one of many variables to consider. It is the mean of all absolute error terms . It can be stated mathematically as:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y - \hat{y}|$$

where N : total number of data points

y : actual values

$\hat{y}$: predicted values

MAPE is a metric measuring how accurate a forecast is. Conventionally, It's measured in percentages. Mathematically :

$$MAPE = \frac{1}{n} \sum_{t=1}^n \left| \frac{A_t - F_t}{A_t} \right|$$

where n : number of observations

t : number of variable terms

A_t : actual value

F_t : forecasted value.

Chapter 5. Time Series Modeling

5.1 Data Collection

The Ganges water level data collection process was done using satellites near a remote village named Chak in Bihar. The location coordinates are:- (25.334,86.175). Low-orbiting satellites outfitted with light-measuring sensors may measure the amount of light being reflected off the bottom, collecting and updating data as they pass by.

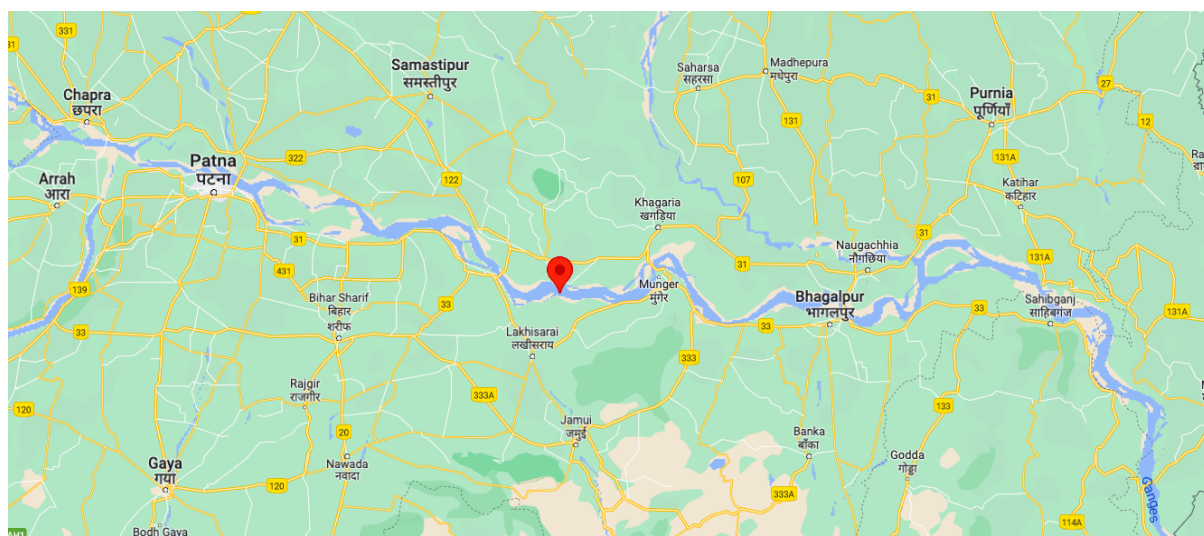


Fig. 3 : Location of the Data Collection point

The dataset contains water level data in meters from 1st January 2000 to 31st December 2013. The length of the dataset was around 5114 with 20% of missing values.

5.2 Data Preprocessing

5.2.1 Imputation of Missing Data

If there are missing values in a time series, there are two options for dealing with the data:

- Keep only the full information-containing record
- Fill up the blanks with the missing information.

There are a lot of methods which are generally used to impute the missing values in a time series. Because time series data contains a time component, only a limited number of statistical approaches are applicable for them. `df.interpolate()` was used to interpolate the missing values. It supports various methods such as 'Linear', 'Time', 'slinear', 'quadratic', 'cubic', 'spline'.

We also used rolling mean and median values of a week to impute the missing values. This is also one of the general approaches widely followed in imputing missing values of a time series.

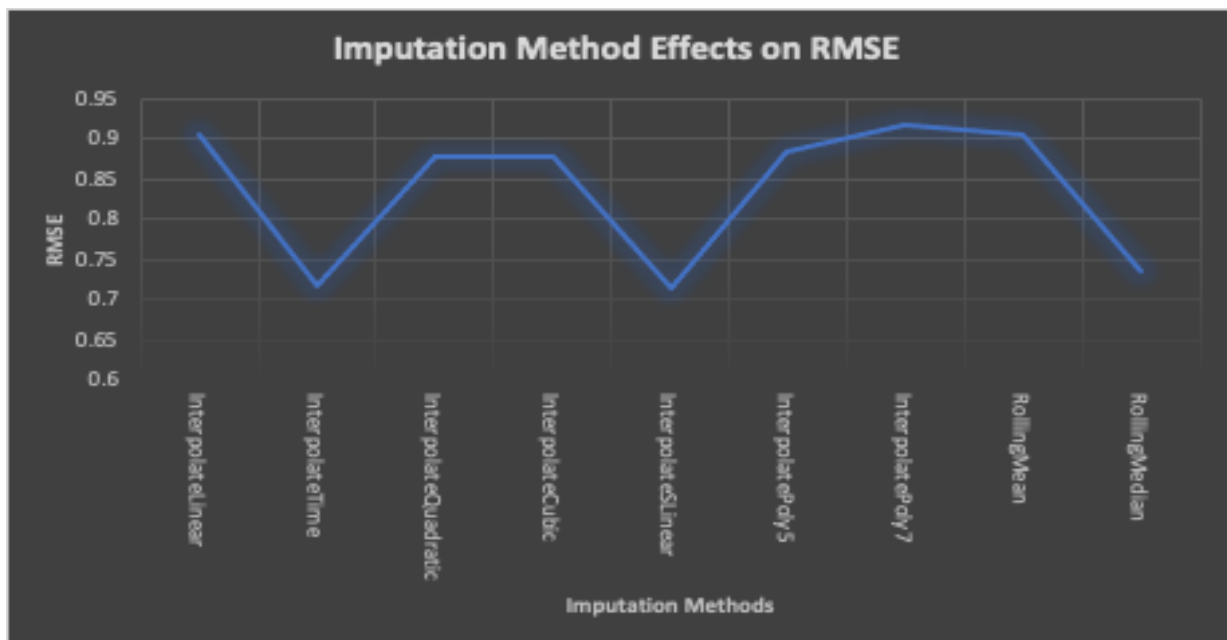


Fig. 4 : RMSE v/s Imputation Effects

The Interpolated dataset obtained after applying each of the above methods was taken to train ARIMA models for forecasting the water level after converting it to a monthly frequency data using mean as a downsampling method. The orders of the

above models were obtained through “Auto-Arima ” function which returns the best ARIMA model according to AIC or BIC values.

We obtained the lowest error for the ‘slinear’ method for the trained model and selected this dataset to carry the further study.

5.2.2 Selecting optimal frequency for the training dataset

We had a given dataset of daily frequency from years 1999 to 2013. It becomes extremely crucial for us to select the data with optimal frequency to train the model for forecast. So, we downsample the daily data to one week, two weeks, three weeks and monthly frequency by taking the average of the values of the water level values in that period. We now trained ARIMA models for each of these downsampled datasets using orders obtained from Auto-ARIMA and calculated the corresponding RMSE values.

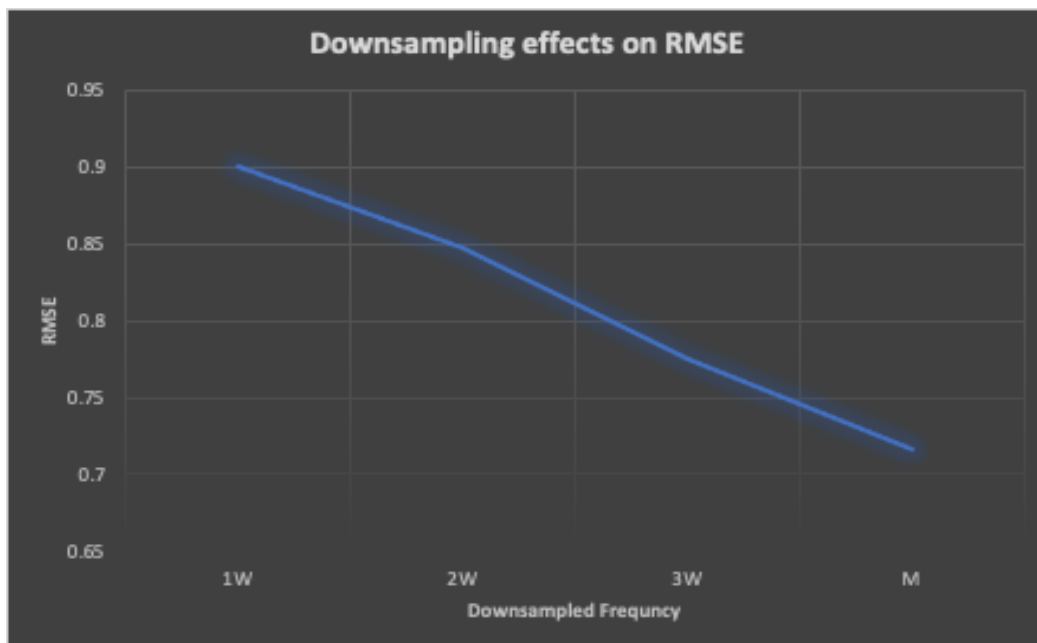


Fig. 5 : Downsampling frequency periods v/s RMSE

The dataset corresponding to lowest RMSE score was selected to carry out the further analysis.

5.2.3 Selecting optimal downsampling method for the training dataset

Earlier we had only taken ‘mean’ as the only downsampling method. There are some other methods such as ‘median’, ‘end’ etc which can be used to downsample the dataset. So it becomes very important for us to select the optimal method and

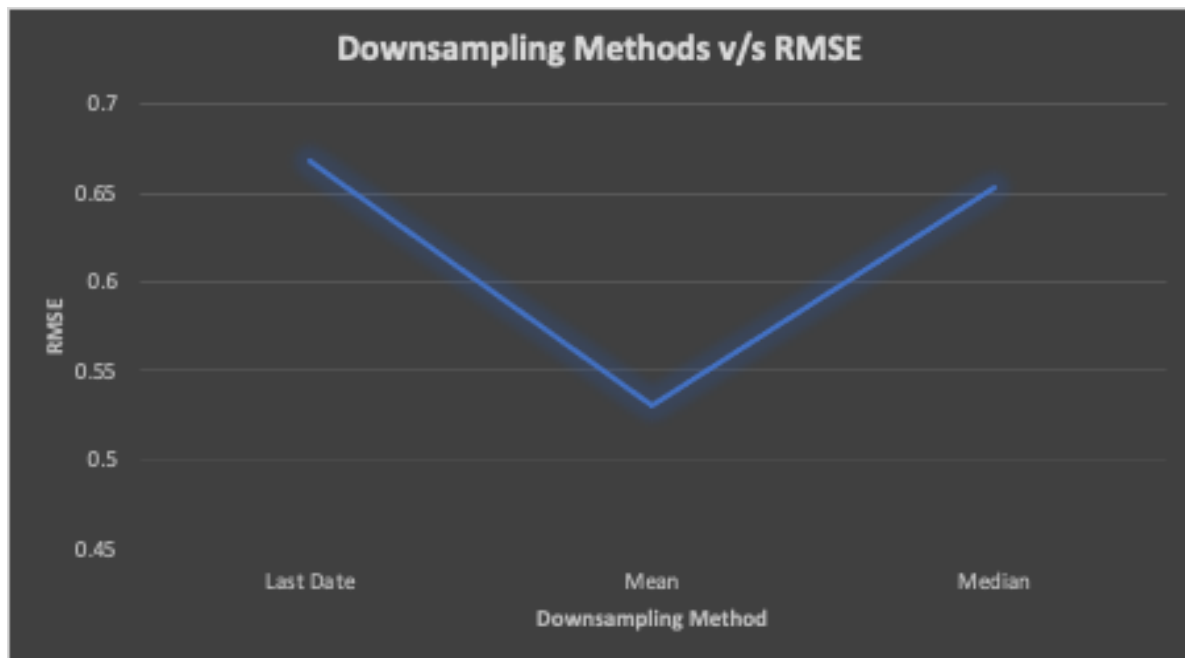


Fig. 6 : Downsampling Methods v/s RMSE

for that we trained three ARIMA models with ‘mean’, ‘median’, ‘end’ methods. We used Auto-ARIMA to select the best order of the models and calculated RMSE scores. The dataset corresponding to lowest RMSE score was selected to carry out the further study.

5.3 Forecasting using ARIMA Model

The selected dataset from above analysis was used to train the final model to forecast the Ganges Water Level. We firstly used ETS decomposition to filter out the Trend, Seasonal and Residual components of our time series.

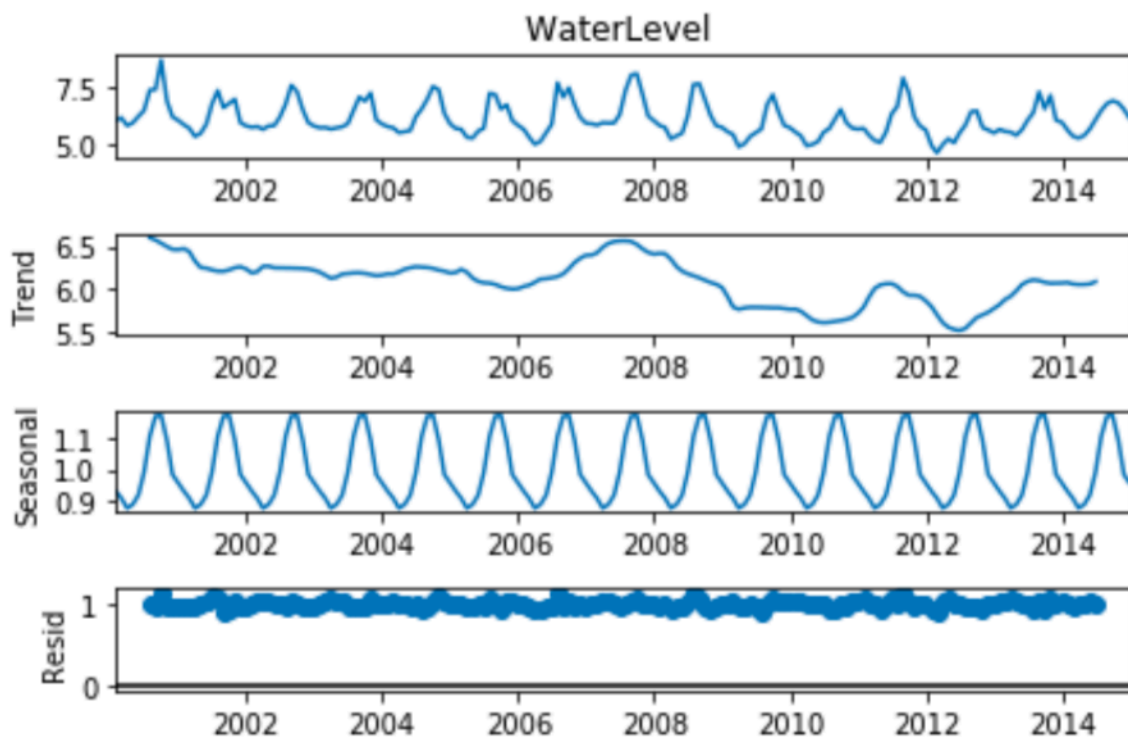
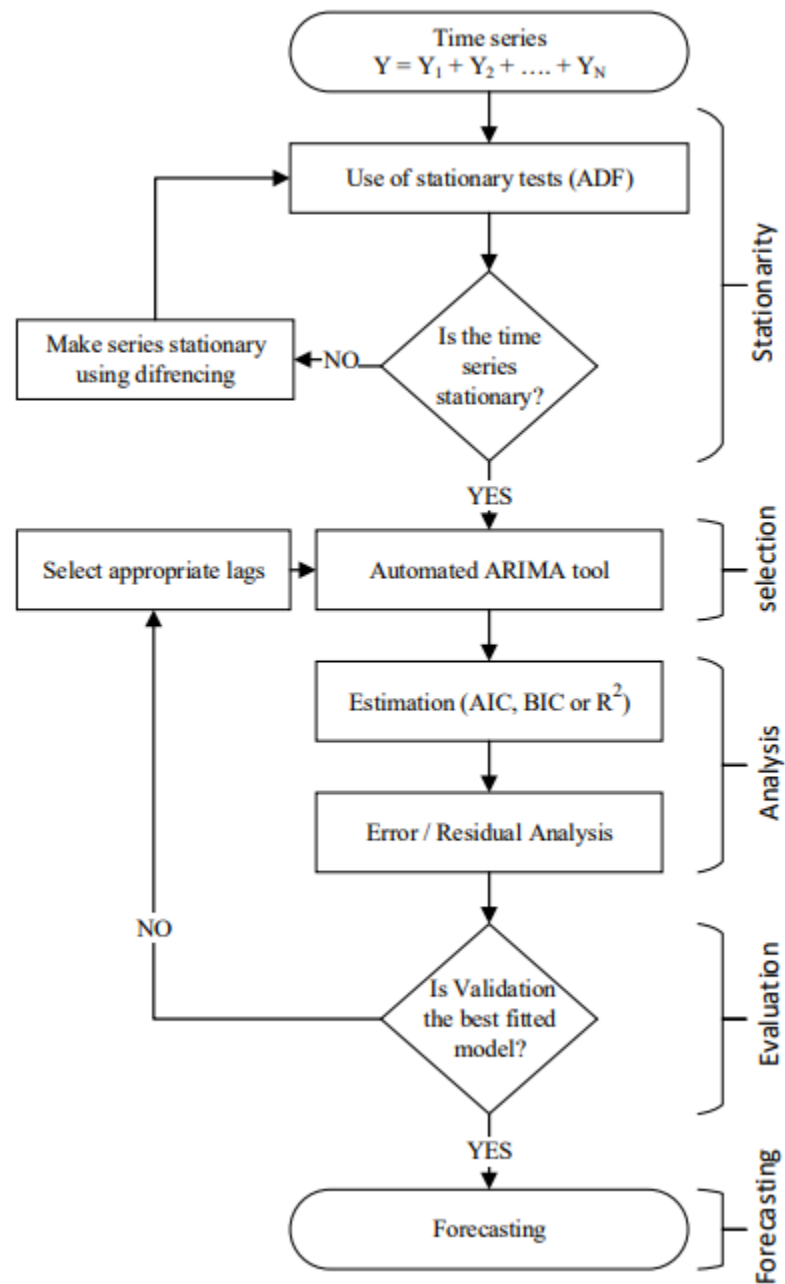


Fig. 7 : ETS decomposition of the WaterLevel time-series

As we can see in above pictures, the time-series has a strong seasonal component which is also evident by the plot of the data too.

We had a total of 168 months of data which we further divided into a training dataset of length of 156 data points and kept 12 months data as our test dataset.

The flowchart shown below depicts standard process of the time series analysis:-



We will first check the stationarity of the time series using the ADF test. The results of the test are shown below which clearly indicates that time series is non-stationary.

Augmented Dickey-Fuller Test:

ADF test statistic -2.337618

p-value 0.160122

lags used 14.000000

observations 153.000000

critical value (1%) -3.473830

critical value (5%) -2.880623

critical value (10%) -2.576945

Weak evidence against the null hypothesis

Fail to reject the null hypothesis

Data has a unit root and is non-stationary

Further to determine ARIMA orders for the best possible model, we used the Auto-ARIMA function. We obtained the lowest AIC value of 183.8 corresponding to ARIMA order of (2,0,3). We used these parameter values to train our final ARIMA model.

```

Performing stepwise search to minimize aic
ARIMA(2,0,2)(0,0,0)[0] : AIC=inf, Time=0.41 sec
ARIMA(0,0,0)(0,0,0)[0] : AIC=1089.249, Time=0.01 sec
ARIMA(1,0,0)(0,0,0)[0] : AIC=inf, Time=0.02 sec
ARIMA(0,0,1)(0,0,0)[0] : AIC=inf, Time=0.14 sec
ARIMA(1,0,1)(0,0,0)[0] : AIC=263.572, Time=0.06 sec
ARIMA(2,0,1)(0,0,0)[0] : AIC=264.738, Time=0.12 sec
ARIMA(1,0,2)(0,0,0)[0] : AIC=262.813, Time=0.23 sec
ARIMA(0,0,2)(0,0,0)[0] : AIC=inf, Time=0.24 sec
ARIMA(1,0,3)(0,0,0)[0] : AIC=263.751, Time=0.12 sec
ARIMA(0,0,3)(0,0,0)[0] : AIC=603.798, Time=0.48 sec
ARIMA(2,0,3)(0,0,0)[0] : AIC=inf, Time=0.54 sec
ARIMA(1,0,2)(0,0,0)[0] intercept : AIC=223.431, Time=0.14 sec
ARIMA(0,0,2)(0,0,0)[0] intercept : AIC=237.208, Time=0.08 sec
ARIMA(1,0,1)(0,0,0)[0] intercept : AIC=231.964, Time=0.11 sec
ARIMA(2,0,2)(0,0,0)[0] intercept : AIC=206.985, Time=0.38 sec
ARIMA(2,0,1)(0,0,0)[0] intercept : AIC=207.209, Time=0.58 sec
ARIMA(3,0,2)(0,0,0)[0] intercept : AIC=209.440, Time=0.62 sec
ARIMA(2,0,3)(0,0,0)[0] intercept : AIC=183.797, Time=0.49 sec
ARIMA(1,0,3)(0,0,0)[0] intercept : AIC=222.160, Time=0.20 sec
ARIMA(3,0,3)(0,0,0)[0] intercept : AIC=210.759, Time=0.87 sec
ARIMA(2,0,4)(0,0,0)[0] intercept : AIC=inf, Time=1.04 sec
ARIMA(1,0,4)(0,0,0)[0] intercept : AIC=226.599, Time=0.89 sec
ARIMA(3,0,4)(0,0,0)[0] intercept : AIC=214.263, Time=1.35 sec

Best model: ARIMA(2,0,3)(0,0,0)[0] intercept
Total fit time: 9.640 seconds

```

Fig. 8 : Summary of the ARIMA models by Auto-ARIMA

5.3 Forecasting using LSTMs

Recurrent neural networks (RNNs) have been predominantly used for sequence modeling for the past couple of decades. Given a time-series data, we can convert it into sequences of numerical values and leverage the strong modeling power of RNNs to predict as well as forecast the future values.

LSTMs are the most common and widely used RNNs in sequence modeling. Therefore, we use an LSTM for our experiments. Our LSTM consists of two hidden layers with 50 output units with ReLU activation in between and a single-layered perceptron on top of it

to obtain the predicted value. We train the LSTM using the ADAM optimizer and on the mean squared error.

We follow a simple mechanism to convert the time-series data into sequences of numerical values. The past n data points are combinedly considered as a single sequence and the data point succeeding these n points is taken as the label.

This way, we have as many training data points for our RNN as the number of data-points in our time-series data minus n .

Here, we experiment with two values, $n=30$ and $n=90$.

Chapter 6. Results

6.1 Summary of ARIMA forecasting

We started our analysis by applying the ARIMA model on the daily dataset but unfortunately the model was not able to capture the pattern in the data very well. We obtained a RMSE score of 1.12 m. So we used average monthly dataset for the forecast.

6.1.1 Parameters of the model

Out of the 168 months of dataset from years 2000 to 2013, 150 observations were taken as training dataset and one year of observations were kept to test the quality of our model.

ARMA Model Results

Dep. Variable:	WaterLevel	No. Observations:	156
Model:	ARMA(2, 3)	Log Likelihood	-78.275
Method:	css-mle	S.D. of innovations	0.388
Date:	Tue, 26 Apr 2022	AIC	170.551
Time:	09:07:27	BIC	191.900
Sample:	01-31-2000	HQIC	179.222
	- 12-31-2012		

	coef	std err	z	P> z	[0.025	0.975]
const	6.1066	0.051	120.289	0.000	6.007	6.206
ar.L1.WaterLevel	1.7304	0.003	605.395	0.000	1.725	1.736
ar.L2.WaterLevel	-0.9994	0.001	-844.012	0.000	-1.002	-0.997
ma.L1.WaterLevel	-1.1795	0.068	-17.449	0.000	-1.312	-1.047
ma.L2.WaterLevel	0.0866	0.102	0.851	0.395	-0.113	0.286
ma.L3.WaterLevel	0.5327	0.061	8.687	0.000	0.413	0.653

Roots

	Real	Imaginary	Modulus	Frequency
AR.1	0.8657	-0.5011j	1.0003	-0.0835
AR.2	0.8657	+0.5011j	1.0003	0.0835
MA.1	0.8568	-0.5163j	1.0003	-0.0863
MA.2	0.8568	+0.5163j	1.0003	0.0863
MA.3	-1.8761	-0.0000j	1.8761	-0.5000

Fig. 9 : Parameters of the ARIMA model

The above picture shows the parameters of the ARIMA model used to forecast the water level.

6.1.2 Comparison of Forecasted and Actual Data

Figure 10 shows a comparison of forecasted and actual data for the ARIMA model. The real data is for the entire year of 2013. The future forecast for one year ahead was generated using this test data as a reference set. Figure 10 shows that actual and anticipated values are very near to each other, with only a few deviating values.

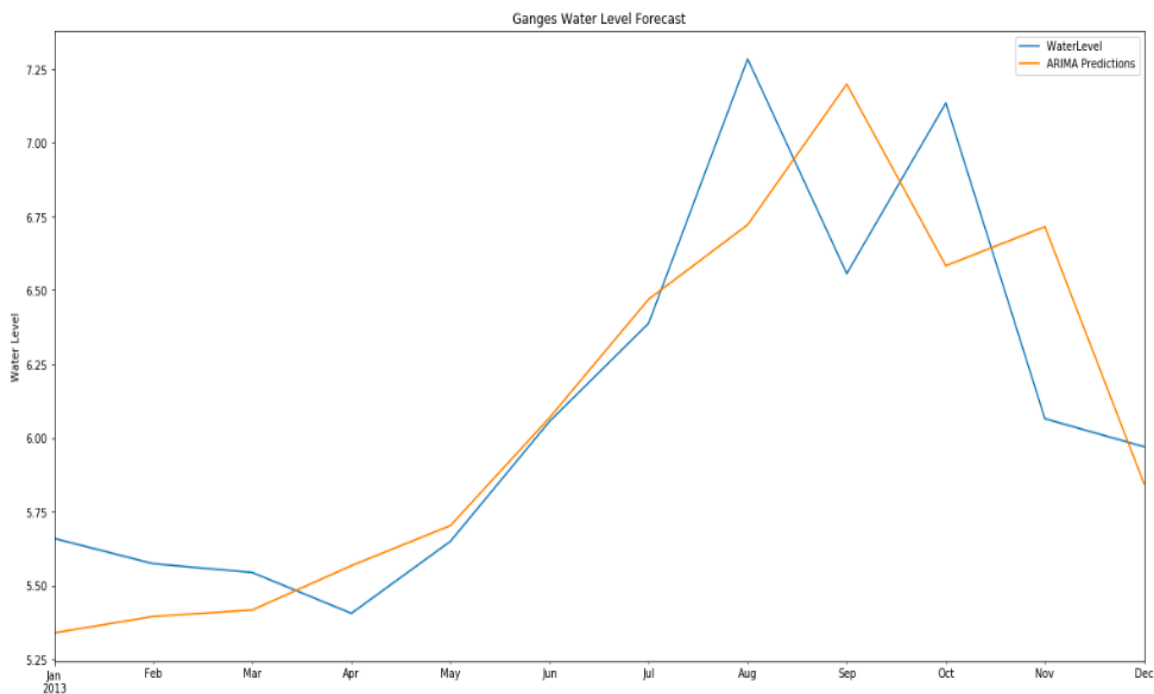


Fig. 10 : Predicted v/s Actual Water Level values using ARIMA model

The anticipated numbers suggest that the river flowed as long as the water level stayed constant owing to glacier melting and weather shifts, and the river basin condition and several other influencing elements remained constant across the forecast timeline. This forecast was created regardless of the weather anomalies that are likely to alter in the future.

The following table contains the details of various error metrics:-

Error Metric	Value
RMSE	0.283 m
MAE	0.263 m
MAPE	4.23%

To build on the findings, this research forecasts statistics for 2014. It can be observed that the historical trend is very similar to the predicted trajectory. The missing data might be linear or nonlinear; nonetheless, for the purposes of study, linear models were used because non-linearity could have considerably influenced the results and varied from the actual river scenario. To solve the anomalies in the river's hydrologic behavior, linear behavior was used to forecast missing data and bring it closer to previously recorded data.

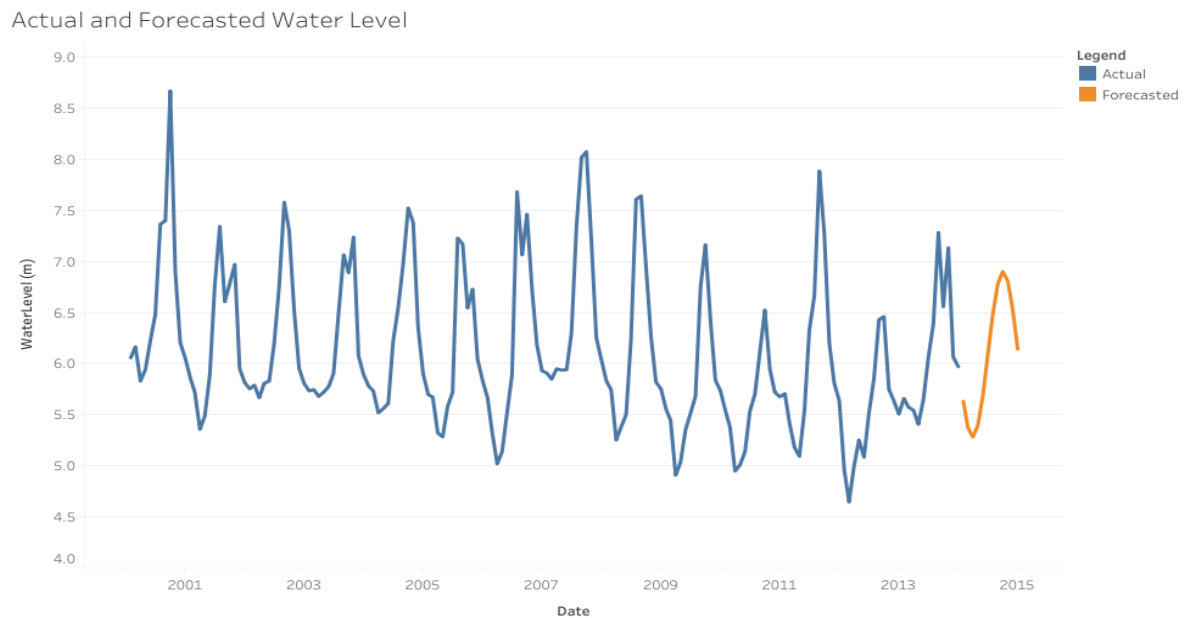


Fig. 11 : Actual and Forecasted values for the Ganges WaterLevel

6.2 Summary of LSTM forecasting

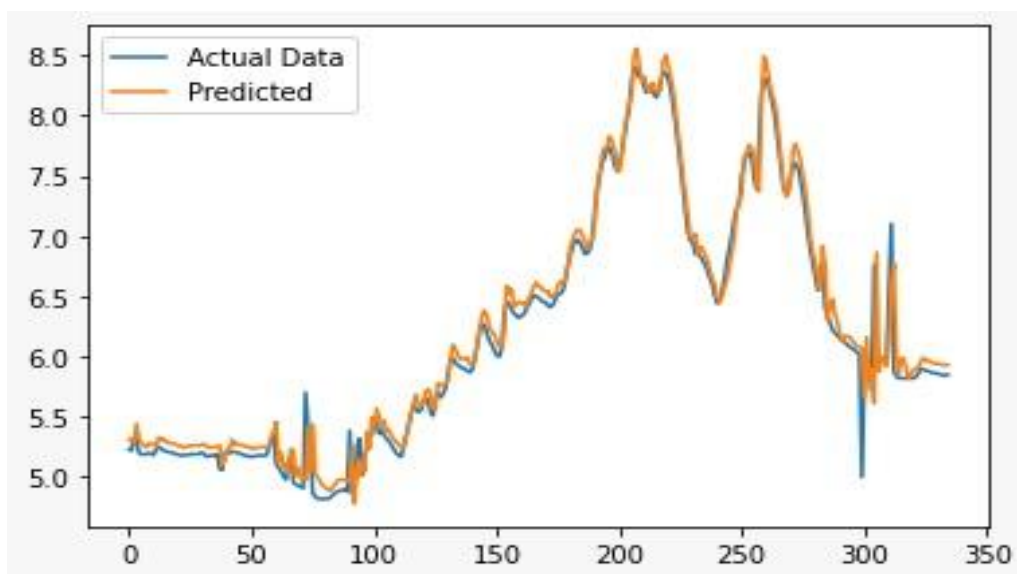
We follow a simple mechanism to convert the time-series data into sequences of numerical values. The past n data points are combinedly considered as a single sequence and the data point succeeding these n points is taken as the label.

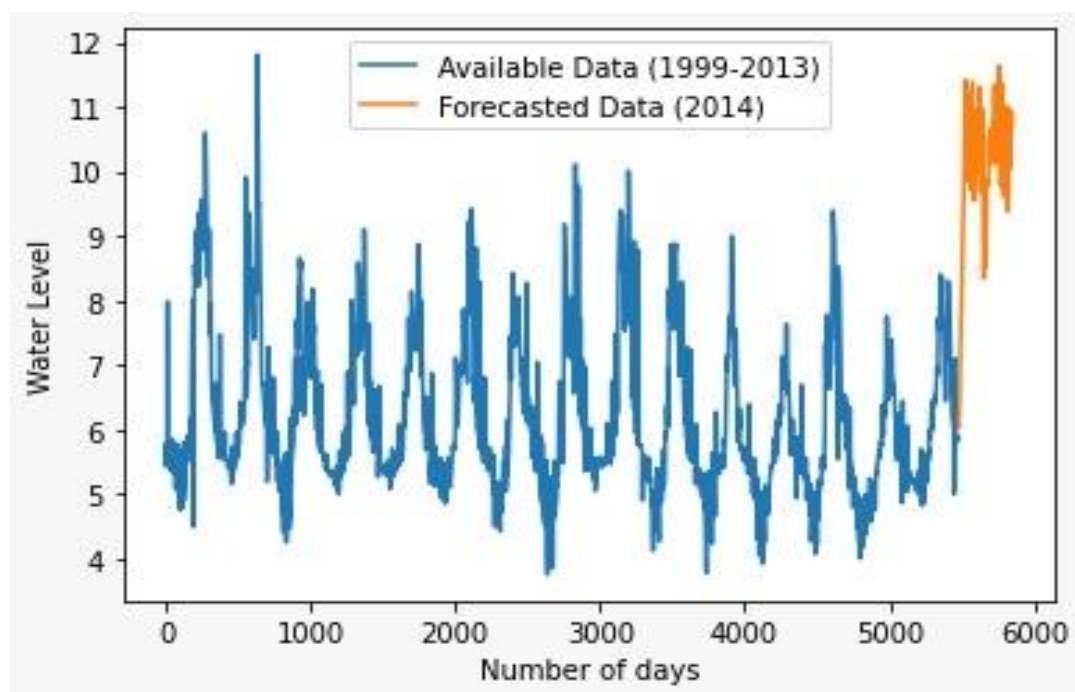
This way, we have as many training data points for our RNN as the number of data-points in our time-series data minus n .

Here, we experiment with two values, $n=30$ and $n=90$.

1) $n=30$

The model does exceptionally well in recognising and identifying the patterns of the test dataset. The RMSE score for this model is around 0.408 m.

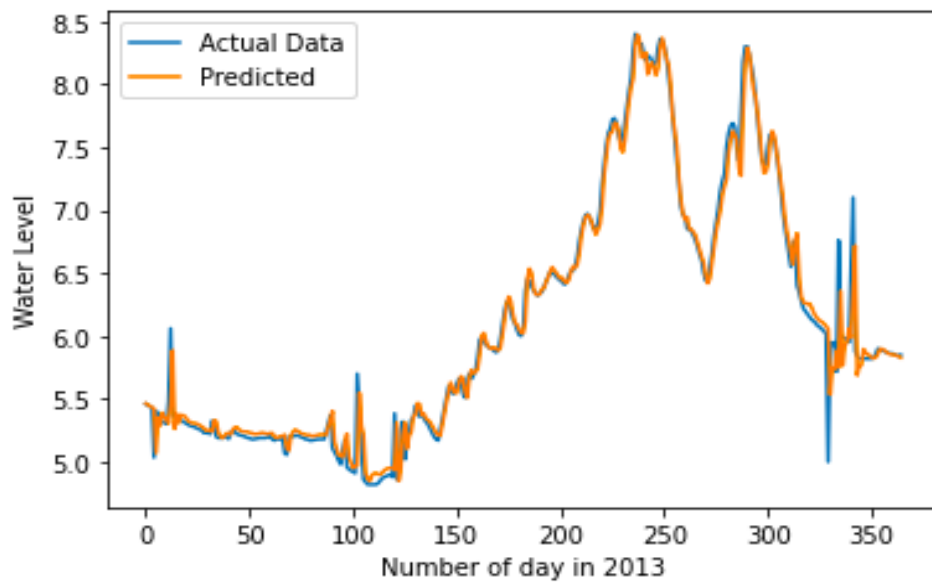




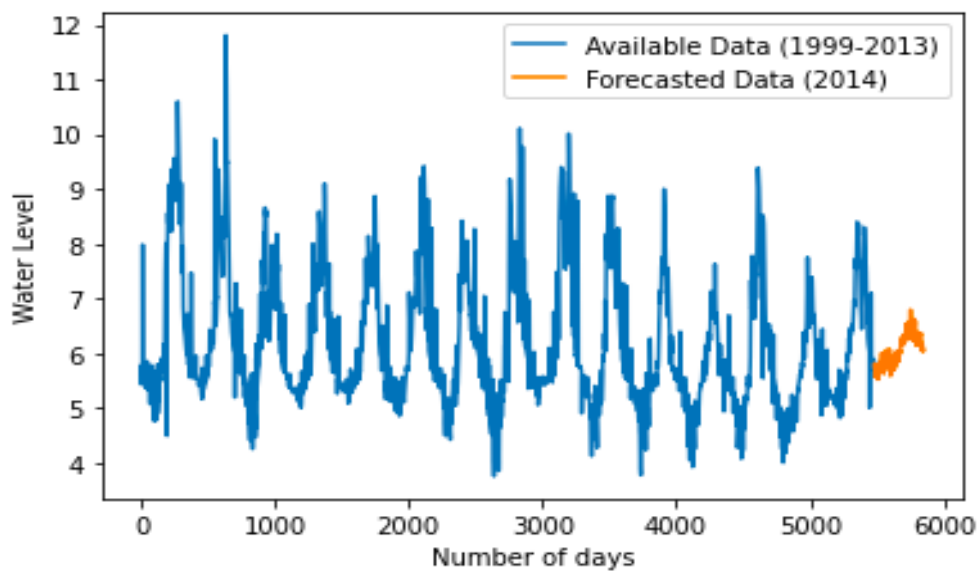
The above plot shows the forecast of the model. Unfortunately, our model hasn't been able to forecast decently. It has been able to capture the trend and seasonality of time series but the magnitude of the patterns hasn't been clearly recognised by the model.

2) **n=90**

We increased the sequence length from 30 to 90 to make the model structure more complex in order to make our model more robust in identifying the patterns in the data to improve our forecast



The RMSE score for the model comes out to be 0.39 m. Though the model has been able to capture the patterns in data and predict reasonably well for the test dataset, it is not able to forecast values in future accurately.



Chapter 7. Discussion

Since the methodology used in this research is auto-forecasting, which does forecasting by selecting the optimal model based on the minimum AIC values, this work can be applied to other domains. Depending upon the linear measurement of previous hydrological data, this tool chose ARIMA (2,0,3). The ARIMA model performs very well on average monthly dataset but for daily frequency data both ARIMA and LSTM are not able to forecast reasonably well. Though LSTM is able to capture and predict the pattern of the test dataset but during the forecast it gets stuck at the local maxima and hence is not able to forecast well. The relevance of this work is that the authorities will be able to use the anticipated results to manage the operation of the new floating barges, which will aid in the seamless delivery of commodities along National Waterway 1.

Despite the use of sophisticated forecasting tools, there will always be an element of uncertainty in the results; as a result, climate quantification is critical for the creation of hydrologic effects. The variability of the outcomes in this study is exactly proportional to the fluctuation in precipitation and temperature throughout time. These two aspects must be appropriately handled in order to increase modeling reliability. Similarly, the inclusion of missing data points could have a significant impact on the model's performance.

Forecasting capability has substantially improved in recent years, and it now has applications in a wide range of sectors. On the other hand, several circumstances contribute to many unexpected findings in the analysis. Unfortunately, these factors are impossible to model or forecast, thus the study must always be taken with a

grain of salt. Precipitation, Temperature, and earthquakes are just some of the examples of factors that might cause significant deviations from predicted values.

Chapter 8. Conclusion

The present study has presented an ARIMA model to forecast average The current study used an ARIMA model to anticipate the Ganges' average monthly water level for the coming year. The models performed well, with an RMSE score of 0.28m for a variable with an average value of roughly 6.11m. The daily water level data of one station, located in rural Bihar, was used to accomplish the objectives. The LSTM was able to capture trends in the test dataset but future forecasts was not very appreciative. Other state-of-the art models can be implemented to maybe improve the accuracy further.

The results can be used by the authorities in charge of the operation of the newly installed floating barges. These findings can be used in a variety of other fields, such as disaster management, flood control, and water resource management.

The results of the analysis could be improved by adding more data in our training dataset which would help the model to capture the intricacies and complex structure of the trend and seasonality of the time series. Other state of the art models such as Transformers can be used to improve the accuracy further. They might be able to capture the daily trends in the time series more effectively.

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